CS506 - Introduction to Quantum Computing

Refined Probability Derivation for Simon's Problem

Scribe: Samuele Perlo

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1 Nomenclature

To define the variables and ensure maximum clarity during the explanation, we introduce the following nomenclature.

We define the vectors as follows:

- $\vec{x}$: A generic vector that can be expressed through its elements as $(x_1, x_2, \ldots, x_n)$.
- $\vec{y}$: Another vector expressed through its elements as $(y_1, y_2, \ldots, y_n)$.

The $\oplus$ operation applied between two vectors is defined as the same operation applied individually to each respective component:

$$\vec{x} \oplus \vec{y} = x_1 \oplus y_1, \quad x_2 \oplus y_2, \quad \ldots \quad x_n \oplus y_n$$

2 Definition of Simon's Problem

2.1 Premises

A function $f : \{0,1\}^n \to X$ is provided. We can assume that the output consists of n bits; in other words, we can set the codomain $X = \{0,1\}^n$ without any loss of information.

2.2 Behavior of the Function

There exists an unknown vector $\vec{s} \in \{0,1\}^n$ such that $\vec{s} \neq 0^n$.

The function f possesses a specific property whereby two outputs coincide, i.e., $f(\vec{x}) = f(\vec{y})$, if and only if one of the following conditions holds for the input variables:

- $\vec{y} = \vec{x}$
- $\vec{y} = \vec{x} \oplus \vec{s}$

2.3 Objective

The objective of the problem is to find the secret vector $\vec{s}$ using the minimum possible number of queries to the function f.

3 Vector Space and Classical Limits

3.1 Definition of the Space Z_2^n

All elements of this space are vectors $\vec{z} \in \{0,1\}^n$. The dimension of this vector space is n.

A fundamental property of this space is that the addition between two vectors modulo 2 is equivalent to the application of the bitwise logical XOR operator:

$$\vec{x} + \vec{y} \pmod 2 = \vec{x} \oplus \vec{y}$$

3.2 Linear Independence and Basis of the Space

Regarding linear independence, a set of vectors $\vec{x}_1, \ldots, \vec{x}_k$ is linearly dependent if there exist k binary values (meaning scalars $\alpha_1, \ldots, \alpha_k \in \{0, 1\}$) not all zero such that their linear combination generates the null vector:

$$\alpha_1 \cdot \vec{x}_1 + \cdots + \alpha_k \cdot \vec{x}_k = 0\ 0 \ldots 0$$

The vectors that constitute the basis of this space of dimension n are those of the canonical basis, having a single bit set to 1 and all others to 0:

$$10 \ldots 00,\ 01 \ldots 00,\ \ldots,\ 00 \ldots 10,\ 00 \ldots 01$$

3.3 Complexity of Classical Algorithms

Simon's problem is intractable to solve efficiently for traditional computers.

Indeed, any classical algorithm that attempts to solve the problem correctly with a probability $\geq \frac{2}{3}$ necessarily requires a number of queries to the function f in the order of $\Omega(2^{n/3})$ queries.

4 Quantum Algorithm and Orthogonal Space

4.1 Quantum Complexity and Operator

Unlike the classical approach, the quantum algorithm is capable of solving the problem more efficiently: $O(n)$ queries are sufficient.

In the example, given the function $f : \{0, 1\}^n \to X$, the action on the quantum oracle is described by the following unitary transformation:

$$U_f|\vec{x}\rangle|\vec{b}\rangle = |\vec{x}\rangle|\vec{b} \oplus \vec{f}(\vec{x})\rangle$$

4.2 Vector Space and Orthogonal Space

Let us consider the vector space S composed of the following elements: $S = \{\vec{0}, \vec{s}\}$.

- $\vec{0}$: vector composed entirely of zeros
- $\vec{s}$: our secret vector

The dimension of this space is $\dim(S) = 1$.

We now define the orthogonal space $S^\perp$ as the set of vectors $\vec{z}$ whose inner product with $\vec{s}$ is zero when evaluated modulo 2:

$$S^\perp = \{\vec{z} \in \{0, 1\}^n \mid \vec{z} \cdot \vec{s} = 0 \pmod 2\}$$

Knowing that the sum of the dimensions of S and its orthogonal complement must coincide with the total dimension of the space n, we obtain:

$$\dim(S) + \dim(S^\perp) = n = \dim(Z_2^n)$$

From this equality we derive that the dimension of the orthogonal space is $\dim(S^\perp) = n - 1$.

4.3 Construction of the System of Equations

The core of the method consists of assuming we can find $n-1$ linearly independent vectors $\vec{z}_1, \vec{z}_2, \ldots, \vec{z}_{n-1}$ belonging to $S^{\perp}$.

If we succeed in this intent, we can set up the following system of linear equations in which the secret vector $\vec{s}$ represents the only unknown variable:

$$\begin{aligned} \vec{z}_1 \cdot \vec{s} &= 0 \pmod 2 \\ \vec{z}_2 \cdot \vec{s} &= 0 \pmod 2 \\ &\vdots \\ \vec{z}_{n-1} \cdot \vec{s} &= 0 \pmod 2 \end{aligned}$$

4.4 Construction and Matrix Form of the System

In this system, the vectors $\vec{z}$ represent the known terms calculated by the algorithm, while the vector $\vec{s}$ represents the unknown we want to find. All elements within this system belong to the binary set $\{0, 1\}$.

This is a system of linear equations modulo 2, meaning evaluated in the field GF(2). We can express it explicitly in matrix form as follows:

$$\begin{pmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n-1,1} & z_{n-1,2} & \cdots & z_{n-1,n} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \pmod 2$$

5 System Resolution and Classical Analysis

5.1 Gaussian Elimination Method

The Gaussian elimination method operates in polynomial time, requiring a time in the order of $O(n^3)$. This procedure yields the vector $\vec{s}' = (s_1, s_2, \ldots, s_n)$ such that $\vec{s}' \neq 0^n$.

5.2 Vector Extraction

We sample randomly and uniformly from the orthogonal space $S^{\perp}$. This operation is repeated $n-1$ times to obtain the vectors $\vec{z}'_1, \vec{z}'_2, \ldots, \vec{z}'_{n-1}$.

We want to calculate the probability that these extracted vectors are linearly independent (a condition we indicate as l.i.):

$$Pr[\vec{z}'_1, \vec{z}'_2, \ldots, \vec{z}'_{n-1} \text{ are l.i.}]$$

This probability can be calculated through the product of the individual conditional probabilities:

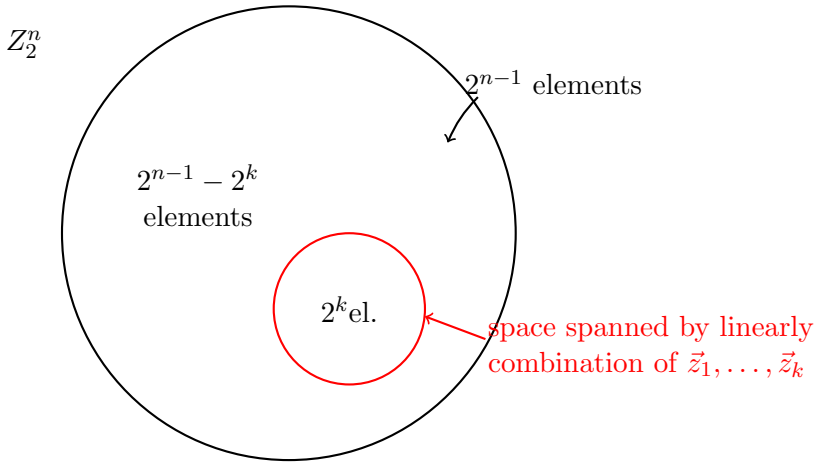
$$\begin{aligned} & Pr[\vec{z}'_1 \text{ is l.i.}] \\ & \times Pr[\vec{z}'_1, \vec{z}'_2 \text{ is l.i.} \mid \vec{z}'_1 \text{ is l.i.}] \\ & \times Pr[\vec{z}'_1, \vec{z}'_2, \vec{z}'_3 \text{ is l.i.} \mid \vec{z}'_1, \vec{z}'_2 \text{ are l.i.}] \\ & \times \ldots \\ & \times Pr[\vec{z}'_1, \ldots, \vec{z}'_{k+1} \text{ is l.i.} \mid \vec{z}'_1, \ldots, \vec{z}'_k \text{ are l.i.}] \\ & \times \ldots \\ & \times Pr[\vec{z}'_1, \ldots, \vec{z}'_{n-1} \text{ is l.i.} \mid \vec{z}'_1, \ldots, \vec{z}'_{n-2} \text{ are l.i.}] \end{aligned}$$

The exact probability of extracting a new vector that is independent from the k previously extracted vectors is defined by the formula:

$$\frac{2^{n-1} - 2^k}{2^{n-1}} = 1 - 2^{k-(n-1)}$$

5.3 Set Representation

Let us consider the entire space Z_2^n which contains 2^{n-1} elements in total. Within it, the space generated by the linear combination of the vectors $\vec{z}_1', \ldots, \vec{z}_k'$ constitutes a subset of 2^k elements. Consequently, the remaining elements useful for drawing a vector that guarantees linear independence are exactly $2^{n-1} - 2^k$.



5.4 Calculation of the Total Probability

Returning to the condition of linear independence, the combination of vectors generates the null vector if the coefficients are zero:

$$\alpha_1\vec{z}_1 + \alpha_2\vec{z}_2 + \cdots + \alpha_k\vec{z}_k = \vec{0} \pmod 2$$

where the coefficients belong to the set $\{0, 1\}$.

The overall probability of extracting $n-1$ linearly independent vectors is given by the following product:

$$\Pr[\vec{z}_1', \ldots, \vec{z}_{n-1}' \text{ sono l.i.}] = \prod_{k=0}^{n-2} \left(1 - 2^{k-(n-1)}\right)$$

Expanding the terms for the various values of k we obtain:

$$\begin{aligned} k = 0 &\to (1 - 2^{-(n-1)})\times \\ k = 1 &\to (1 - 2^{1-(n-1)}) \times \ldots \\ k = n-2 &\to (1 - 2^{(n-2)-(n-1)}) \end{aligned}$$

This can also be written in the equivalent detailed product-form as shown in the derivation:

$$= 1 \cdot \left(1 - \frac{2^1}{2^{n-1}}\right)\left(1 - \frac{2^2}{2^{n-1}}\right)\cdots\left(1 - \frac{2^{n-2}}{2^{n-1}}\right)$$

To solve this product, we use an inequality valid for binomials:

$$
\begin{aligned}
(1-a_1)(1-a_2) &= 1-a_1-a_2+a_1a_2 > 1-(a_1+a_2) \\
(1-a_1)(1-a_2)(1-a_3) &> 1-(a_1+a_2+a_3) \\
&\vdots \\
(1-a_1)\cdots(1-a_r) &> 1-(a_1+\cdots+a_r)
\end{aligned}
$$

Applying the inequality to our product, we can determine the lower-bound by evaluating the geometric series:

$$
\begin{aligned}
\prod_{k=0}^{n-2}(1-2^{k-(n-1)}) &= (1-2^{-(n-1)})(1-2^{1-(n-1)})\cdots(1-2^{-1}) \\
&> 1-\sum_{j=1}^{n-1}2^{-j}
\end{aligned}
$$

Refining this through the detailed summation of the inner terms yields:

$$
\begin{aligned}
\Pr[\vec{z}_1',\ldots,\vec{z}_{m-1}' \text{ are l.i.}] &= \left(1-\frac{2^1}{2^{m-1}}\right)\left(1-\frac{2^2}{2^{m-1}}\right)\cdots\left(1-\frac{2^{m-2}}{2^{m-1}}\right) \\
&> \frac{1}{2}\left(1-\left(\frac{2^1}{2^{m-1}}+\frac{2^2}{2^{m-1}}+\cdots+\frac{2^{m-3}}{2^{m-1}}\right)\right) \\
&> \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}
\end{aligned}
$$

Side note: As $n \to \infty$, the limit of this q-series converges to approximately $0.28879\ldots$, which safely satisfies the bound being strictly greater than one-quarter.